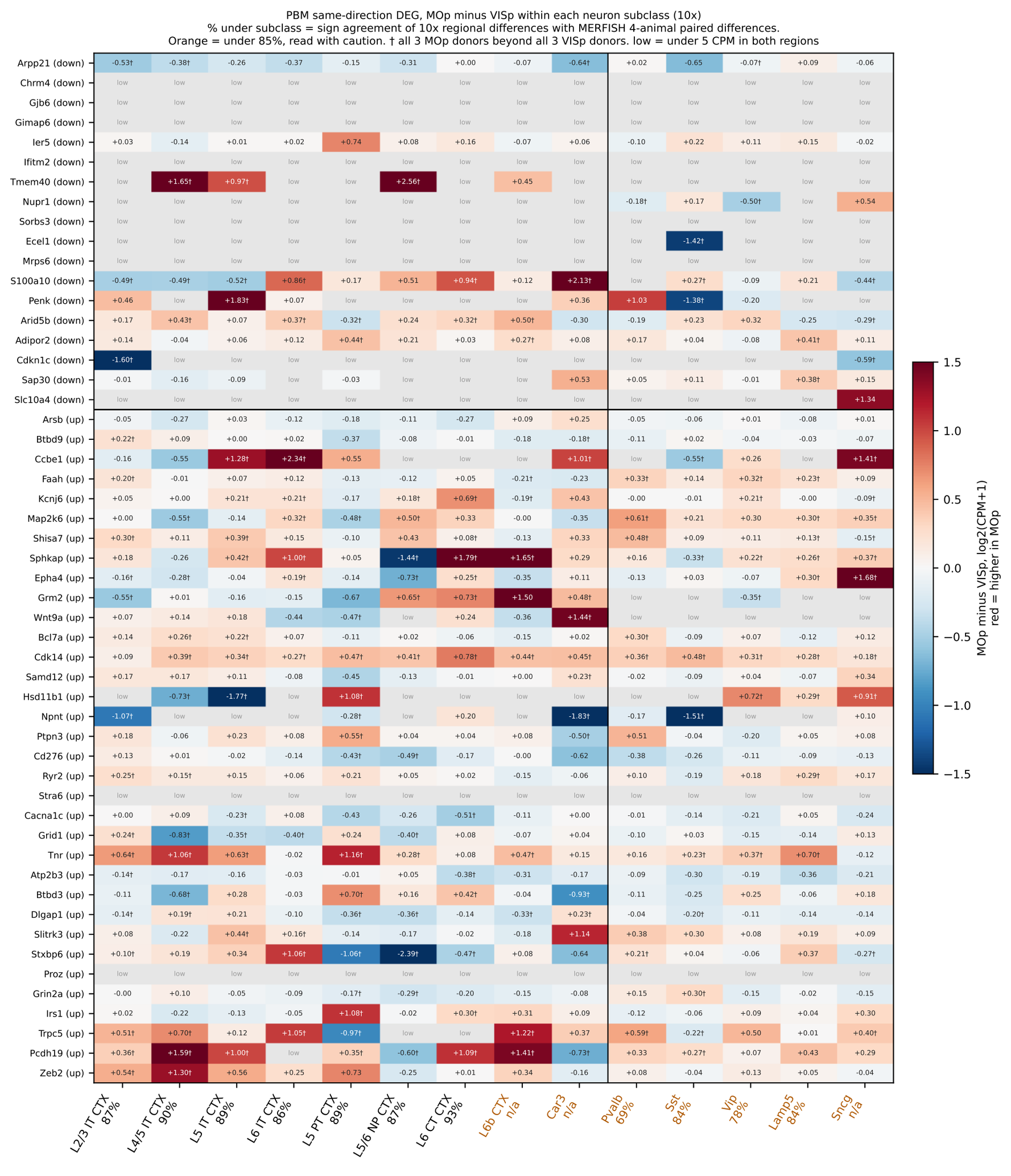
PBM same-direction DEG, MOp minus VISp within each neuron subclass (10x)
% under subclass = sign agreement of 10x regional differences with MERFISH 4-animal paired differences.
Orange = under 85%, read with caution. † all 3 MOp donors beyond all 3 VISp donors. low = under 5 CPM in both regions
Arpp21 (down) -0.53† -0.38† -0.26 -0.37 -0.15 -0.31 +0.00 -0.07 -0.64† +0.02 -0.65 -0.07† +0.09 -0.06
Chrm4 (down) low low low low low low low low low low low low low low
Gjb6 (down) low low low low low low low low low low low low low low
Gimap6 (down) low low low low low low low low low low low low low low
Ier5 (down) +0.03 -0.14 +0.01 +0.02 +0.74 +0.08 +0.16 -0.07 +0.06 -0.10 +0.22 +0.11 +0.15 -0.02
Ifitm2 (down) low low low low low low low low low low low low low low
Tmem40 (down) low +1.65† +0.97† low low +2.56† low +0.45 low low low low low low
Nupr1 (down) low low low low low low low low low -0.18† +0.17 -0.50† low +0.54
Sorbs3 (down) low low low low low low low low low low low low low low
Ecel1 (down) low low low low low low low low low low -1.42† low low low
Mrps6 (down) low low low low low low low low low low low low low low
S100a10 (down) -0.49† -0.49† -0.52† +0.86† +0.17 +0.51 +0.94† +0.12 +2.13† low +0.27† -0.09 +0.21 -0.44†
Penk (down) +0.46 low +1.83† +0.07 low low low low +0.36 +1.03 -1.38† -0.20 low low
Arid5b (down) +0.17 +0.43† +0.07 +0.37† -0.32† +0.24 +0.32† +0.50† -0.30 -0.19 +0.23 +0.32 -0.25 -0.29†
Adipor2 (down) +0.14 -0.04 +0.06 +0.12 +0.44† +0.21 +0.03 +0.27† +0.08 +0.17 +0.04 -0.08 +0.41† +0.11
Cdkn1c (down) -1.60† low low low low low low low low low low low low -0.59†
Sap30 (down) -0.01 -0.16 -0.09 low -0.03 low low low +0.53 +0.05 +0.11 -0.01 +0.38† +0.15
Slc10a4 (down) low low low low low low low low low low low low low +1.34
Arsb (up) -0.05 -0.27 +0.03 -0.12 -0.18 -0.11 -0.27 +0.09 +0.25 -0.05 -0.06 +0.01 -0.08 +0.01
Btbd9 (up) +0.22† +0.09 +0.00 +0.02 -0.37 -0.08 -0.01 -0.18 -0.18† -0.11 +0.02 -0.04 -0.03 -0.07
Ccbe1 (up) -0.16 -0.55 +1.28† +2.34† +0.55 low low low +1.01† low -0.55† +0.26 low +1.41†
Faah (up) +0.20† -0.01 +0.07 +0.12 -0.13 -0.12 +0.05 -0.21† -0.23 +0.33† +0.14 +0.32† +0.23† +0.09
Kcnj6 (up) +0.05 +0.00 +0.21† +0.21† -0.17 +0.18† +0.69† -0.19† +0.43 -0.00 -0.01 +0.21† -0.00 -0.09†
Map2k6 (up) +0.00 -0.55† -0.14 +0.32† -0.48† +0.50† +0.33 -0.00 -0.35 +0.61† +0.21 +0.30 +0.30† +0.35†
Shisa7 (up) +0.30† +0.11 +0.39† +0.15 -0.10 +0.43 +0.08† -0.13 +0.33 +0.48† +0.09 +0.11 +0.13† -0.15†
Sphkap (up) +0.18 -0.26 +0.42† +1.00† +0.05 -1.44† +1.79† +1.65† +0.29 +0.16 -0.33† +0.22† +0.26† +0.37†
Epha4 (up) -0.16† -0.28† -0.04 +0.19† -0.14 -0.73† +0.25† -0.35 +0.11 -0.13 +0.03 -0.07 +0.30† +1.68†
Grm2 (up) -0.55† +0.01 -0.16 -0.15 -0.67 +0.65† +0.73† +1.50 +0.48† low low -0.35† low low
Wnt9a (up) +0.07 +0.14 +0.18 -0.44 -0.47† low +0.24 -0.36 +1.44† low low low low low
Bcl7a (up) +0.14 +0.26† +0.22† +0.07 -0.11 +0.02 -0.06 -0.15 +0.02 +0.30† -0.09 +0.07 -0.12 +0.12
Cdk14 (up) +0.09 +0.39† +0.34† +0.27† +0.47† +0.41† +0.78† +0.44† +0.45† +0.36† +0.48† +0.31† +0.28† +0.18†
Samd12 (up) +0.17 +0.17 +0.11 -0.08 -0.45 -0.13 -0.01 +0.00 +0.23† -0.02 -0.09 +0.04 -0.07 +0.34
Hsd11b1 (up) low -0.73† -1.77† low +1.08† low low low low low low +0.72† +0.29† +0.91†
Npnt (up) -1.07† low low low -0.28† low +0.20 low -1.83† -0.17 -1.51† low low +0.10
Ptpn3 (up) +0.18 -0.06 +0.23 +0.08 +0.55† +0.04 +0.04 +0.08 -0.50† +0.51 -0.04 -0.20 +0.05 +0.08
Cd276 (up) +0.13 +0.01 -0.02 -0.14 -0.43† -0.49† -0.17 -0.00 -0.62 -0.38 -0.26 -0.11 -0.09 -0.13
Ryr2 (up) +0.25† +0.15† +0.15 +0.06 +0.21 +0.05 +0.02 -0.15 -0.06 +0.10 -0.19 +0.18 +0.29† +0.17
Stra6 (up) low low low low low low low low low low low low low low
Cacna1c (up) +0.00 +0.09 -0.23† +0.08 -0.43 -0.26 -0.51† -0.11 +0.00 -0.01 -0.14 -0.21 +0.05 -0.24
Grid1 (up) +0.24† -0.83† -0.35† -0.40† +0.24 -0.40† +0.08 -0.07 +0.04 -0.10 +0.03 -0.15 -0.14 +0.13
Tnr (up) +0.64† +1.06† +0.63† -0.02 +1.16† +0.28† +0.08 +0.47† +0.15 +0.16 +0.23† +0.37† +0.70† -0.12
Atp2b3 (up) -0.14† -0.17 -0.16 -0.03 -0.01 +0.05 -0.38† -0.31 -0.17 -0.09 -0.30 -0.19 -0.36 -0.21
Btbd3 (up) -0.11 -0.68† +0.28 -0.03 +0.70† +0.16 +0.42† -0.04 -0.93† -0.11 -0.25 +0.25 -0.06 +0.18
Dlgap1 (up) -0.14† +0.19† +0.21 -0.10 -0.36† -0.36† -0.14 -0.33† +0.23† -0.04 -0.20† -0.11 -0.14 -0.14
Slitrk3 (up) +0.08 -0.22 +0.44† +0.16† -0.14 -0.17 -0.02 -0.18 +1.14 +0.38 +0.30 +0.08 +0.19 +0.09
Stxbp6 (up) +0.10† +0.19 +0.34 +1.06† -1.06† -2.39† -0.47† +0.08 -0.64 +0.21† +0.04 -0.06 +0.37 -0.27†
Proz (up) low low low low low low low low low low low low low low
Grin2a (up) -0.00 +0.10 -0.05 -0.09 -0.17† -0.29† -0.20 -0.15 -0.08 +0.15 +0.30† -0.15 -0.02 -0.15
Irs1 (up) +0.02 -0.22 -0.13 -0.05 +1.08† -0.02 +0.30† +0.31 +0.09 -0.12 -0.06 +0.09 +0.04 +0.30
Trpc5 (up) +0.51† +0.70† +0.12 +1.05† -0.97† low low +1.22† +0.37 +0.59† -0.22† +0.50 +0.01 +0.40†
Pcdh19 (up) +0.36† +1.59† +1.00† low +0.35† -0.60† +1.09† +1.41† -0.73† +0.33 +0.27† +0.07 +0.43 +0.29
Zeb2 (up) +0.54† +1.30† +0.56 +0.25 +0.73 -0.25 +0.01 +0.34 -0.16 +0.08 -0.04 +0.13 +0.05 -0.04
L2/3 IT CTX 87%
L4/5 IT CTX 90%
L5 IT CTX 89%
L6 IT CTX 86%
L5 PT CTX 89%
L5/6 NP CTX 87%
L6 CT CTX 93%
L6b CTX n/a
Car3 n/a
Pvalb 69%
Sst 84%
Vip 78%
Lamp5 84%
Sncg n/a
MOp minus VISp, log2(CPM+1)
red = higher in MOp
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5